QUICK REFERENCE

VC

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

VCF Command Reference

Quick-reference guide to essential CLI commands for VCF component management, troubleshooting, and daily operations.

CLI COMMANDS | MANAGEMENT | TROUBLESHOOTING | DAILY OPS

VCF 9.0 VMware Cloud Foundation

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VCF 9.0.1 Command Reference

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1. SDDC Manager API

Get Auth Token

```
TOKEN=$(curl -sk -X POST https://sddc-manager.lab.local/v1/tokens \
  -H "Content-Type: application/json" \
  -d '{"username":"administrator@vsphere.local","password":"<pass>"}' \
  | python3 -c "import sys,json;print(json.load(sys.stdin)['accessToken'])")
```

Core Endpoints

```
# System health
curl -sk -H "Authorization: Bearer $TOKEN" https://localhost/v1/system/health

# NSX cluster status (check for ACTIVE vs ACTIVATING)
curl -sk -H "Authorization: Bearer $TOKEN" https://localhost/v1/nsxt-clusters |
python3 -m json.tool

# Resource locks (stale locks block all operations)
curl -sk -H "Authorization: Bearer $TOKEN" https://localhost/v1/resource-locks |
python3 -m json.tool

# Credentials
curl -sk -H "Authorization: Bearer $TOKEN" https://localhost/v1/credentials |
python3 -m json.tool

# Task details (UI hides error payloads)
curl -sk -H "Authorization: Bearer $TOKEN" https://localhost/v1/tasks/<task-id> |
python3 -m json.tool

# System notifications
curl -sk -H "Authorization: Bearer $TOKEN"
https://localhost/v1/system/notifications

# Domains
curl -sk -H "Authorization: Bearer $TOKEN" https://localhost/v1/domains | python3
-m json.tool

# Hosts
curl -sk -H "Authorization: Bearer $TOKEN" https://localhost/v1/hosts | python3 -
m json.tool

# Failed tasks (newest first)
curl -sk -H "Authorization: Bearer $TOKEN" "https://localhost/v1/tasks?
status=FAILED&pageSize=5&sortOrder=DESC" | python3 -m json.tool
```

```
# Stuck IN_PROGRESS tasks (these block new operations)
curl -sk -H "Authorization: Bearer $TOKEN" "https://localhost/v1/tasks?
status=IN_PROGRESS" | python3 -c \
  "import sys,json; d=json.load(sys.stdin); print(f'Count:
{len(d.get(\"elements\",[]))}'); [print(f'  {t[\"type\"]} |
{t[\"creationTimestamp\"]}') for t in d.get('elements',[])]"

# NSX credentials
curl -sk -H "Authorization: Bearer $TOKEN" "https://localhost/v1/credentials?
resourceType=NSXT_MANAGER" | python3 -m json.tool

# Cancel stuck task (NOTE: API often rejects with TA_TASK_CAN_NOT_BE_RETRIED — use
DB fix instead)
curl -sk -X PATCH https://localhost/v1/tasks/<task-id> \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{"status":"CANCELLED"}'
```

---

2. SDDC Manager Services & Logs

Service Management

```
# Check all VCF services
systemctl status domainmanager lcm operationsmanager commonsvcs postgresql nginx

# Restart all services (recommended method)
/opt/vmware/vcf/operationsmanager/scripts/cli/sddcmanager_restart_services.sh

# Restart individual services
systemctl restart domainmanager
systemctl restart lcm
systemctl restart operationsmanager
systemctl restart commonsvcs
systemctl restart sddc-manager-ui-app.service

# List all VCF service units
systemctl list-units --type=service | grep vcf
```

Key Log Locations

```
# Domain manager (deployment, bringup)
tail -100 /var/log/vmware/vcf/domainmanager/domainmanager.log

# Operations manager (credential rotation, tasks)
tail -100 /var/log/vmware/vcf/operationsmanager/operationsmanager.log

# Lifecycle manager (bundles, upgrades, certs)
tail -100 /var/log/vmware/vcf/lcm/lcm.log
tail -100 /var/log/vmware/vcf/lcm/lcm-debug.log

# Common services
tail -100 /var/log/vmware/vcf/commonsvcs/commonsvcs.log
```

Log Search

```
# Search for errors
grep -ri "error\|exception\|failed"
/var/log/vmware/vcf/domainmanager/domainmanager.log | tail -50

# Search for cert/TLS issues
grep -i "tlsfatal\|ssl\|certificate" /var/log/vmware/vcf/lcm/lcm-debug.log | tail
-20

# Search by timestamp
grep "2026-02-22 14:" /var/log/vmware/vcf/operationsmanager/operationsmanager.log

# Follow log in real time
tail -f /var/log/vmware/vcf/domainmanager/domainmanager.log

# Search by task ID
grep "<task-id>" /var/log/vmware/vcf/lcm/lcm.log

# Journalctl
journalctl -u vcf-services --since "1 hour ago" -p err
```

Service Properties Ownership Fix

```
chown vcf:vcf /opt/vmware/vcf/domainmanager/conf/application-prod.properties
chown vcf:vcf /opt/vmware/vcf/operationsmanager/conf/application-prod.properties
```

NFS Mount Ownership Fix

```
chown root:vcf /nfs/vmware/vcf/nfs-mount/
```

---

3. SDDC Manager PostgreSQL Database

Enable Database Access (Trust Auth Workaround)

```
# Backup pg_hba.conf
cp /data/pgdata/pg_hba.conf /data/pgdata/pg_hba.conf.bak

# Set trust auth (temporary — ALWAYS restore after)
sed -i 's/scram-sha-256/trust/g' /data/pgdata/pg_hba.conf
su - postgres -c "/usr/pgsql/15/bin/pg_ctl reload -D /data/pgdata"
```

Connect to Databases

```
# IMPORTANT: Always disable psql pager to prevent --More-- prompts
export PAGER=cat
export PGPAGER=cat

# Platform database (resource states, locks, task metadata)
su - postgres -c "PAGER=cat psql -h 127.0.0.1 -d platform"

# Operations manager database (tasks, executions)
su - postgres -c "PAGER=cat psql -h 127.0.0.1 -d operationsmanager"

# List all databases
su - postgres -c "PAGER=cat psql -h 127.0.0.1 -t -c \"SELECT datname FROM
pg_database WHERE datistemplate = false;\""
```

```
# Returns: postgres, lcm, platform, operationsmanager, domainmanager,
sddc_manager_ui

# List tables in a database
su - postgres -c "PAGER=cat psql -h 127.0.0.1 -d platform -t -c \"SELECT tablename
FROM pg_tables WHERE schemaname = 'public' ORDER BY tablename;\""
```

Credential Cascade Failure — Full Repair

```
# ── Step 1: Fix NSX resource status (platform DB) ──
# Check current status (should be ACTIVE, may be ACTIVATING or ERROR)
su - postgres -c "PAGER=cat psql -h 127.0.0.1 -d platform -t -c \"SELECT id,
status FROM nsxt;\""

# Fix ANY non-ACTIVE status
su - postgres -c "PAGER=cat psql -h 127.0.0.1 -d platform -c \"UPDATE nsxt SET
status = 'ACTIVE' WHERE status != 'ACTIVE';\""

# ── Step 2: Clear stale resource locks (platform DB) ──
su - postgres -c "PAGER=cat psql -h 127.0.0.1 -d platform -c \"SELECT count(*)
FROM lock;\""
su - postgres -c "PAGER=cat psql -h 127.0.0.1 -d platform -c \"DELETE FROM
lock;\""

# ── Step 3: Mark stuck tasks as resolved (platform DB) ──
su - postgres -c "PAGER=cat psql -h 127.0.0.1 -d platform -c \"SELECT resolved,
count(*) FROM task_metadata GROUP BY resolved;\""
su - postgres -c "PAGER=cat psql -h 127.0.0.1 -d platform -c \"UPDATE
task_metadata SET resolved = true WHERE resolved = false;\""

# ── Step 4: Clear task locks (platform DB) ──
su - postgres -c "PAGER=cat psql -h 127.0.0.1 -d platform -c \"DELETE FROM
task_lock;\""
```

Key tables: `platform` *DB:* `nsxt` *(status),* `lock` *,* `task_metadata` *(resolved),* `task_lock` *.* `operationsmanager` *DB:* `task` *(column:* `state` *, not* `status` *),* `execution` *(column:* `execution_status` *),* `processing_task` *,* `execution_to_task` *.*

Restore pg_hba.conf (CRITICAL — Do Immediately)

```
cp /data/pgdata/pg_hba.conf.bak /data/pgdata/pg_hba.conf
su - postgres -c "/usr/pgsql/15/bin/pg_ctl reload -D /data/pgdata"

# Verify restored (should return 4+)
grep -c 'scram-sha-256' /data/pgdata/pg_hba.conf
```

Restart After DB Fix

```
systemctl restart operationsmanager
systemctl is-active operationsmanager
```

---

4. SDDC Manager SSH & Account Recovery

SSH Access

```
# Only vcf user can SSH (root/admin rejected)
ssh vcf@192.168.1.241

# Escalate to root
su -

# File transfer (SCP doesn't work — restricted shell)
ssh vcf@192.168.1.241 "cat > /home/vcf/myfile.zip" < localfile.zip
ssh vcf@192.168.1.241 "cat /path/to/remote/file" > local_copy
```

Account Lockout (faillock)

```
# Check lockout status
faillock --user vcf

# Reset locked accounts (run from console as root)
```

```
faillock --user vcf --reset
faillock --user root --reset
```

---

5. NSX Manager API

Cluster & Node Status

```
# Cluster health (via VIP)
curl -sk -u admin:'<pass>' https://192.168.1.70/api/v1/cluster/status

# Cluster health (via node — use when VIP is down)
curl -sk -u admin:'<pass>' https://192.168.1.71/api/v1/cluster/status

# Cluster info + node UUIDs
curl -sk -u admin:'<pass>' https://192.168.1.71/api/v1/cluster
```

Transport & Networking

```
# Transport node status
curl -sk -u admin:'<pass>' https://192.168.1.70/api/v1/transport-nodes/state

# Transport zones
curl -sk -u admin:'<pass>' https://192.168.1.70/api/v1/transport-zones

# Segments
curl -sk -u admin:'<pass>' https://192.168.1.70/policy/api/v1/infra/segments

# Compute managers
curl -sk -u admin:'<pass>' https://192.168.1.71/api/v1/fabric/compute-managers
```

Service Health

```
# Individual service status
curl -sk -u admin:'<pass>' https://192.168.1.71/api/v1/node/services/http/status
curl -sk -u admin:'<pass>' https://192.168.1.71/api/v1/node/services/search/status
```

```
# Node system status (includes load average)
curl -sk -u admin:'<pass>' https://192.168.1.71/api/v1/node/status
```

Certificate Management

```
# List all certificates
curl -sk -u admin:'<pass>' https://192.168.1.71/api/v1/trust-
management/certificates

# Import certificate (see Section 20 for full workflow)
curl -sk -u admin:'<pass>' -X POST \
  "https://192.168.1.71/api/v1/trust-management/certificates?action=import" \
  -H "Content-Type: application/json" -d @/tmp/nsx-import.json
```

---

6. NSX Admin CLI

SSH as `admin` user (not root):

```
get cluster status                    # Cluster health
get managers                          # List manager nodes
get name-servers                      # DNS config
set name-servers <ip>                 # Set DNS (NOT via UI)
get ntp-servers                       # NTP config
set ntp-servers <ip>                  # Set NTP (NOT via UI)
```

---

7. NSX Node Services

SSH as `root` user:

```
# Service status
systemctl status proton               # Main NSX Manager API
systemctl status search               # Search service
systemctl status nsx-appl-proxy       # Appliance proxy
systemctl status nsx-sha              # System health agent
```

```
systemctl status corfu_server        # Distributed datastore

# Service restart
systemctl restart proton
systemctl restart search
systemctl restart nsx-appl-proxy

# Load monitoring (nested: load >30 = still booting)
cat /proc/loadavg
ps -eo pid,%cpu,%mem,comm --sort=-%cpu | head -15

# Disk space
df -h
```

---

8. vCenter Server Management

Service Control

```
# Status
service-control --status              # All services summary
service-control --status --all        # All services detailed
service-control --status vpxd         # Specific service

# Start / Stop / Restart
service-control --start --all
service-control --stop vpxd
service-control --restart vpxd
service-control --restart --all

# Alternative service listing
vmon-cli --list
vmon-cli --status vpxd
```

Certificate Manager

```
/usr/lib/vmware-vmca/bin/certificate-manager
```

```
# List certificate stores
for store in MACHINE_SSL_CERT TRUSTED_ROOTS; do
  /usr/lib/vmware-vmafd/bin/vecs-cli entry list --store $store
done
```

Database

```
# Check PostgreSQL
service-control --status vmware-vpostgres

# Connect
/opt/vmware/vpostgres/current/bin/psql -U postgres

# Active connections
/opt/vmware/vpostgres/current/bin/psql -U postgres -c "SELECT count(*) FROM
pg_stat_activity;"
```

Log Locations

```
# VPXD (main vCenter service)
tail -100 /var/log/vmware/vpxd/vpxd.log
grep -i "error\|exception" /var/log/vmware/vpxd/vpxd.log | tail -50

# UI
tail -100 /var/log/vmware/vsphere-ui/logs/vsphere_client_virgo.log

# First boot (during deployment)
cat /var/log/firstboot/firstbootStatus.json
tail -50 /var/log/vmware/firstboot/installer.log
```

Appliance Info

```
vamicli version --appliance
vamicli update --check
```

---

9. vCenter / vSphere API

```
# Create session
SESSION=$(curl -sk -X POST https://vcenter.lab.local/api/session \
  -u 'administrator@vsphere.local:<pass>' | tr -d '"')

# List VMs
curl -sk -H "vmware-api-session-id: $SESSION"
https://vcenter.lab.local/api/vcenter/vm

# MOB (browser-based deep object access)
# https://vcenter.lab.local/mob
```

---

10. ESXi Host — System

```
esxcli system hostname get                              # Show FQDN
esxcli system hostname set --fqdn=esxi01.lab.local      # Set FQDN
esxcli system version get                                # ESXi version/build
esxcli system time get                                   # System time
esxcli system maintenanceMode get                        # Check maintenance mode
esxcli system maintenanceMode set -e true -m noAction    # Enter (no data
evacuation)
esxcli system maintenanceMode set -e true -m evacuateAllData  # Enter (vSAN
evacuate)
esxcli system maintenanceMode set -e false               # Exit maintenance mode
esxcli system ssh set --enable=true                      # Enable SSH
esxcli system ssh get                                    # SSH status
esxcli hardware cpu global get                           # CPU info (EVC check)
```

---

11. ESXi Host — Network

```
# VMkernel adapters
esxcli network ip interface list
```

```
esxcli network ip interface ipv4 get -i vmk0
esxcli network ip interface add -i vmk1 -p "vMotion"
esxcli network ip interface remove -i vmk1

# Routing
esxcli network ip route ipv4 list
esxcli network ip neighbor list

# vSwitch
esxcli network vswitch standard list
esxcli network vswitch dvs vmware list

# Physical NICs
esxcli network nic list
esxcli network nic get -n vmnic0
esxcli network nic stats get -n vmnic0

# Connections / Firewall
esxcli network ip connection list
esxcli network ip connection list | grep 1234
esxcli network firewall ruleset list
esxcli network firewall ruleset list | grep -i ssh

# vmkping
vmkping 192.168.1.75                          # Basic ping
vmkping -I vmk2 192.168.1.75                  # Ping from specific vmk
vmkping -d -s 1572 192.168.1.75               # Jumbo frame test

# esxcfg
esxcfg-vmknic -l                              # VMkernel NIC list
esxcfg-vswitch -l                             # vSwitch list
esxcfg-nics -l                                # Physical NIC list
```

---

12. ESXi Host — Storage

```
# Device listing
esxcli storage core device list
esxcli storage core device list | grep -E "Display Name|Is SSD"
esxcli storage core adapter list
esxcli storage filesystem list
```

```
esxcli storage vmfs extent list

# Rescan
esxcli storage core adapter rescan --all

# SSD detection (force for nested)
esxcli storage nmp satp rule add -s VMW_SATP_LOCAL -d <device> -o enable_ssd
esxcli storage core claiming reclaim -d <device>
esxcli storage nmp satp rule list | grep enable_ssd

# Disk eligibility for vSAN
vdq -qH
vdq -q -d <device>

# Partition management
partedUtil getptbl /vmfs/devices/disks/<device>
partedUtil delete /vmfs/devices/disks/<device> 1
partedUtil delete /vmfs/devices/disks/<device> 2
```

---

13. ESXi Host — VM Management

```
# List VMs
vim-cmd vmsvc/getallvms

# Power operations
vim-cmd vmsvc/power.getstate <vmid>
vim-cmd vmsvc/power.on <vmid>
vim-cmd vmsvc/power.off <vmid>
vim-cmd vmsvc/power.shutdown <vmid>
vim-cmd vmsvc/power.reset <vmid>

# Register / Unregister
vim-cmd solo/registervm "/vmfs/volumes/<datastore>/<vm>/<vm>.vmx"
vim-cmd vmsvc/unregister <vmid>

# Device info
vim-cmd vmsvc/device.getdevices <vmid>

# BIOS boot
vim-cmd vmsvc/setboot.options <vmid> enterBIOSSetup=true
```

```
# Hardware info
vim-cmd hostsvc/hosthardware

# Maintenance mode
vim-cmd hostsvc/maintenance_mode_enter
vim-cmd hostsvc/maintenance_mode_exit
```

---

14. ESXi Host — Services & Troubleshooting

Service Management

```
# Host services
/etc/init.d/hostd restart
/etc/init.d/hostd status
/etc/init.d/vpxa restart
/etc/init.d/vpxa status
services.sh restart                          # Restart all host services

# SSH
/etc/init.d/SSH start
/etc/init.d/SSH stop
/etc/init.d/SSH status
vim-cmd hostsvc/enable_ssh && vim-cmd hostsvc/start_ssh

# NSX services on host
/etc/init.d/nsx-proxy status
/etc/init.d/nsx-proxy restart
/etc/init.d/nsx-datapath status
/etc/init.d/nsx-opsagent status

# Auto-backup
/sbin/auto-backup.sh
```

Certificate Regeneration

```
mv /etc/vmware/ssl/rui.crt /etc/vmware/ssl/rui.crt.bak
mv /etc/vmware/ssl/rui.key /etc/vmware/ssl/rui.key.bak
/sbin/generate-certificates
services.sh restart
openssl x509 -in /etc/vmware/ssl/rui.crt -text -noout | grep -A1 "Subject
Alternative Name"
```

Get Host Thumbprints

```
echo | openssl s_client -connect 192.168.1.74:443 2>/dev/null | openssl x509 -
noout -fingerprint -sha256
echo | openssl s_client -connect 192.168.1.75:443 2>/dev/null | openssl x509 -
noout -fingerprint -sha256
echo | openssl s_client -connect 192.168.1.76:443 2>/dev/null | openssl x509 -
noout -fingerprint -sha256
echo | openssl s_client -connect 192.168.1.82:443 2>/dev/null | openssl x509 -
noout -fingerprint -sha256
```

Performance / Diagnostics

```
esxtop                                        # Interactive performance monitor
esxtop -b -d 5 -n 10 > /tmp/esxtop.csv       # Batch mode to CSV
```

vMotion Fix (vhv.enable Ghost Setting)

```
# Diagnose
grep -i vhv /vmfs/volumes/<datastore>/<vm>/<vm>.vmx

# Fix (removing the line is NOT enough — must set explicitly)
echo 'vhv.enable = "FALSE"' >> /vmfs/volumes/<datastore>/<vm>/<vm>.vmx
```

NSX Host Logs

```
tail -50 /var/log/nsx-syslog.log
esxcli network ip connection list | grep 1234
esxcfg-vmknic -l | grep -i tep
```

---

15. vSAN Operations

Health & Status

```
esxcli vsan health cluster list              # Health summary
esxcli vsan health cluster get               # Detailed health
esxcli vsan health cluster get -t "Network health"
esxcli vsan cluster get                      # Cluster info
esxcli vsan network list                     # vSAN network config
esxcli vsan debug resync summary get         # Resync status
esxcli vsan debug object health summary get  # Object health
esxcli vsan debug object list                # All objects
```

Storage & Disk Groups

```
esxcli vsan storage list                     # Disk/storage status
esxcli vsan storage automode set --enabled=false
esxcli vsan storage automode set --enabled=true
esxcli vsan storage add -s <cache> -d <capacity>
esxcli vsan storage remove -s <device>
```

Network

```
esxcli vsan network ip add -i vmk1
esxcli vsan network ip remove -i vmk1
vmkping -I vmk2 192.168.12.120               # TEP-to-TEP connectivity
```

Logging

```
grep -i "vsan\|cmmds\|clom\|dom\|lsom" /var/log/vmkernel.log | tail -50
tail -50 /var/log/vsanmgmt.log
tail -50 /var/log/vsantraced.log
/etc/init.d/vsanmgmtd status
/etc/init.d/vsand status
```

vSAN Observer (RVC)

```
rvc administrator@vsphere.local@localhost
cd /vcenter.lab.local/vcenter-dc01/computers/vcenter-cl01
vsan.observer . --run-webserver --force
```

---

16. vSAN Disk Management

Remove Disk from vSAN & Clean Partitions

```
esxcli vsan storage remove -d <device>
partedUtil getptbl /vmfs/devices/disks/<device>
partedUtil delete /vmfs/devices/disks/<device> 1
partedUtil delete /vmfs/devices/disks/<device> 2
vdq -q                                      # Verify eligible again
```

---

17. vmkfstools — Disk Cloning & Management

```
# Clone thick to thin
vmkfstools -i <source>.vmdk <dest>.vmdk -d thin

# Clone thick to thick
vmkfstools -i <source>.vmdk <dest>.vmdk -d zeroedthick
vmkfstools -i <source>.vmdk <dest>.vmdk -d eagerzeroedthick
```

```
# Create new disk
vmkfstools -c 50G -d thin /vmfs/volumes/<ds>/<vm>/newdisk.vmdk

# Extend disk
vmkfstools -X 100G /vmfs/volumes/<ds>/<vm>/disk.vmdk

# Delete disk
vmkfstools -U /vmfs/volumes/<ds>/<vm>/disk.vmdk

# Get disk geometry
vmkfstools -g /vmfs/volumes/<ds>/<vm>/disk.vmdk

# Disk descriptor info
vmkfstools -D "/vmfs/volumes/<ds>/<vm>/<vm>.vmdk"
```

---

18. Certificate Operations — OpenSSL

Generate Self-Signed Certificate

```
openssl req -x509 -newkey rsa:2048 -keyout server.key -out server.crt \
  -days 365 -nodes -subj "/CN=hostname"
```

Generate with SANs

```
openssl req -x509 -newkey rsa:2048 -keyout server.key -out server.crt \
  -days 365 -nodes -subj "/CN=192.168.1.52/O=VCF-Depot/C=US" \
  -addext "subjectAltName=IP:192.168.1.52,DNS:localhost" \
  -addext "keyUsage=digitalSignature,keyEncipherment" \
  -addext "extendedKeyUsage=serverAuth"
```

Generate CSR

```
openssl genrsa -out server.key 2048
```

```
openssl req -new -key server.key -out server.csr -subj "/CN=hostname/O=Org/C=US"
```

View Certificate Details

```
openssl x509 -in cert.crt -text -noout                              # Full details
openssl x509 -in cert.crt -noout -subject                           # Subject only
openssl x509 -in cert.crt -noout -issuer                            # Issuer only
openssl x509 -in cert.crt -noout -dates                             # Valid dates
openssl x509 -in cert.crt -noout -enddate                           # Expiry only
openssl x509 -in cert.crt -text -noout | grep -A2 "Subject Alternative"  # SANs
openssl x509 -in cert.crt -noout -fingerprint -sha256               # Fingerprint
```

Check Remote Certificate

```
# View remote cert
echo | openssl s_client -connect <host>:443 2>/dev/null | openssl x509 -noout -
text | grep -A2 "Subject Alternative"

# Download remote cert to file
openssl s_client -showcerts -connect <host>:443 < /dev/null 2>/dev/null | openssl
x509 -outform PEM > /tmp/cert.pem

# Test TLS connectivity
openssl s_client -connect <host>:<port>
openssl s_client -connect <host>:<port> -tls1_2
openssl s_client -connect <host>:<port> -tls1_2 </dev/null 2>&1 | grep -E
"Cipher|Protocol|Verify"
```

Convert Formats

```
openssl x509 -in cert.pem -outform der -out cert.der               # PEM → DER
openssl x509 -in cert.der -inform der -outform pem -out cert.pem   # DER → PEM
```

Verify Certificate

```
openssl verify -CAfile ca.crt server.crt
```

---

19. Certificate Operations — Keytool

List Certificates

```
keytool -list -keystore /etc/alternatives/jre/lib/security/cacerts -storepass
changeit
keytool -list -alias <alias> -keystore /etc/alternatives/jre/lib/security/cacerts
-storepass changeit -v
```

Import Certificate

```
# Into Java cacerts
keytool -importcert -alias <alias> -file /tmp/cert.crt \
  -keystore /etc/alternatives/jre/lib/security/cacerts \
  -storepass changeit -noprompt

# Into VCF trust store
KEY=$(cat /etc/vmware/vcf/commonsvcs/trusted_certificates.key)
keytool -importcert -alias <alias> -file /tmp/cert.crt \
  -keystore /etc/vmware/vcf/commonsvcs/trusted_certificates.store \
  -storepass "$KEY" -noprompt
```

Delete Certificate

```
keytool -delete -alias <alias> \
  -keystore /etc/alternatives/jre/lib/security/cacerts -storepass changeit
```

Find All Cacerts on System

```
find / -name "cacerts" -type f 2>/dev/null
```

Restart After Keystore Changes

```
/opt/vmware/vcf/operationsmanager/scripts/cli/sddcmanager_restart_services.sh
```

---

20. NSX Certificate Replacement (Full Workflow)

Step 1 — Create OpenSSL Config

```
cat > /tmp/nsx-cert.conf << 'EOF'
[ req ]
default_bits = 2048
distinguished_name = req_distinguished_name
req_extensions = req_ext
x509_extensions = req_ext
prompt = no

[ req_distinguished_name ]
countryName = US
stateOrProvinceName = Lab
localityName = Lab
organizationName = lab.local
commonName = nsx-vip.lab.local

[ req_ext ]
basicConstraints = CA:FALSE
subjectAltName = @alt_names

[alt_names]
DNS.1 = nsx-vip.lab.local
DNS.2 = nsx-node1.lab.local
DNS.3 = nsx-manager.lab.local
IP.1 = 192.168.1.70
```

```
IP.2 = 192.168.1.71
EOF
```

Step 2 — Generate Certificate

```
openssl req -x509 -nodes -days 825 -newkey rsa:2048 \
  -keyout /tmp/nsx.key -out /tmp/nsx.crt \
  -config /tmp/nsx-cert.conf -sha256
```

Step 3 — Verify SANs

```
openssl x509 -in /tmp/nsx.crt -text -noout | grep -A4 "Subject Alternative Name"
```

Step 4 — Build JSON Payload

```
python -c "
import json
cert = open('/tmp/nsx.crt').read()
key = open('/tmp/nsx.key').read()
print(json.dumps({'pem_encoded': cert, 'private_key': key}))
" > /tmp/nsx-import.json
```

Step 5 — Import via NSX API

```
curl -sk -u admin:'<pass>' -X POST \
  "https://192.168.1.71/api/v1/trust-management/certificates?action=import" \
  -H "Content-Type: application/json" -d @/tmp/nsx-import.json
```

Step 6 — Get Node UUID

```
curl -sk -u admin:'<pass>' https://192.168.1.71/api/v1/cluster
```

Step 7 — Apply to Node

```
curl -sk -u admin:'<pass>' -X POST \
  "https://192.168.1.71/api/v1/trust-management/certificates/<cert-id>?
action=apply_certificate&service_type=API&node_id=<node-uuid>"
```

Step 8 — Apply to VIP

```
curl -sk -u admin:'<pass>' -X POST \
  "https://192.168.1.71/api/v1/trust-management/certificates/<cert-id>?
action=apply_certificate&service_type=MGMT_CLUSTER"
```

Step 9 — Verify

```
openssl s_client -connect 192.168.1.71:443 -showcerts </dev/null 2>/dev/null |
openssl x509 -noout -text | grep -A2 "Subject Alternative Name"
openssl s_client -connect 192.168.1.70:443 -showcerts </dev/null 2>/dev/null |
openssl x509 -noout -text | grep -A2 "Subject Alternative Name"
```

Step 10 — Import into SDDC Manager Trust Stores

```
# Pull active NSX cert
openssl s_client -showcerts -connect 192.168.1.71:443 < /dev/null 2>/dev/null |
openssl x509 -outform PEM > /tmp/nsx-root.crt

# VCF trust store
KEY=$(cat /etc/vmware/vcf/commonsvcs/trusted_certificates.key)
keytool -importcert -alias nsx-selfsigned -file /tmp/nsx-root.crt \
  -keystore /etc/vmware/vcf/commonsvcs/trusted_certificates.store \
  -storepass "$KEY" -noprompt

# Java cacerts
keytool -importcert -alias nsx-selfsigned -file /tmp/nsx-root.crt \
  -keystore /etc/alternatives/jre/lib/security/cacerts \
  -storepass changeit -noprompt
```

```
# Restart services
/opt/vmware/vcf/operationsmanager/scripts/cli/sddcmanager_restart_services.sh
```

---

21. VCF Operations for Logs — Certificate

Create OpenSSL Config

```
cat > /tmp/vrli-cert.cnf << 'EOF'
[req]
default_bits = 4096
prompt = no
default_md = sha256
distinguished_name = dn
req_extensions = v3_req
x509_extensions = v3_req

[dn]
C = US
ST = California
L = Lab
O = Lab
OU = VCF
CN = logs.lab.local

[v3_req]
basicConstraints = CA:FALSE
keyUsage = digitalSignature, keyEncipherment
extendedKeyUsage = serverAuth, clientAuth
subjectAltName = @alt_names

[alt_names]
DNS.1 = logs.lab.local
DNS.2 = logs
IP.1 = 192.168.1.242
EOF
```

Generate and Verify

```
openssl req -x509 -nodes -days 730 -newkey rsa:4096 \
  -keyout /tmp/vrli.key -out /tmp/vrli.crt -config /tmp/vrli-cert.cnf

openssl x509 -in /tmp/vrli.crt -noout -text | grep -A5 "Subject Alternative Name"
cat /tmp/vrli.crt
cat /tmp/vrli.key
```

---

22. Offline Depot — Server & Bundles

Windows — Start HTTPS Depot Server

```
cd C:\VCF-DEPOT
python https_server.py
```

Windows — Generate Certificate

```
cd C:\VCF-DEPOT
python generate_cert.py
```

Windows — Firewall Rule

```
netsh advfirewall firewall add rule name="VCF Depot 8443" dir=in action=allow
protocol=tcp localport=8443
```

Windows — Extract Metadata

```
Expand-Archive -Path "vcf-9.0.1.0-offline-depot-metadata.zip" -DestinationPath
"C:\VCF-Depot\metadata-extract" -Force
Copy-Item "C:\VCF-Depot\metadata-extract\PROD\*" "C:\VCF-Depot\PROD\" -Recurse -
Force
```

SDDC Manager — Import Depot Certificate

```
# Download cert from depot server
openssl s_client -connect 192.168.1.160:8443 </dev/null 2>/dev/null | openssl x509
-outform PEM > /tmp/depot.crt

# Import into Java truststore
CACERTS=$(find /usr -name cacerts 2>/dev/null | head -1)
keytool -importcert -trustcacerts -alias offline-depot -file /tmp/depot.crt \
  -keystore $CACERTS -storepass changeit -noprompt

# Verify
keytool -list -alias offline-depot -keystore $CACERTS -storepass changeit

# Restart LCM
systemctl restart lcm
```

SDDC Manager — Verify Depot Connectivity

```
curl -k -u admin:admin
https://192.168.1.160:8443/PROD/COMP/SDDC_MANAGER_VCF/lcm/productVersionCatalog/prod
```

---

23. SOS Diagnostic Bundle

```
cd /opt/vmware/sddc-support/

# Collect log bundle
./sos --domain-name mgmt --log-bundle

# Log bundle with health check
./sos --domain-name mgmt --log-bundle --health-check

# Health check only
./sos --health-check

# Get inventory
```

```
./sos --get-inventory

# Get all passwords
./sos --get-passwords

# Backup config
./sos --backup-config

# Cleanup old logs
./sos --cleanup-logs
```

---

24. VDT (VMware Depot Tool)

```
# Upload to SDDC Manager
ssh vcf@192.168.1.241 "cat > /home/vcf/vdt.zip" < vdt-2.2.7_02-05-2026.zip

# Extract and run
su -
cd /home/vcf
unzip vdt-2.2.7_02-05-2026.zip
cd vdt-2.2.7_02-05-2026
python vdt.py
# Prompted for administrator@vsphere.local password
```

> ***Note:*** *VDT is NOT pre-installed. Download from Broadcom KB 344917.*

---

25. ovftool — Manual OVA Deployment

Discover vCenter Inventory Path

```
ovftool --noSSLVerify "vi://administrator%40vsphere.local:
<pass>@vcenter.lab.local/"
```

Deploy OVA

```
ovftool --acceptAllEulas --skipManifestCheck --noSSLVerify \
  --diskMode=thin --powerOn \
  --name=<vm-name> \
  --datastore=vcenter-cl01-ds-vsan01 \
  --net:"Network 1"="vcenter-cl01-vds01-pg-vm-mgmt" \
  --prop:vamitimezone=UTC \
  --prop:root_password='<pass>' \
  "/path/to/appliance.ova" \
  "vi://administrator%40vsphere.local:<url-encoded-
pass>@vcenter.lab.local/vcenter-dc01/host/vcenter-cl01/"
```

Note: *Use single-line commands. Backslash continuation breaks* `--noSSLVerify` .

---

26. Windows Host — DNS & Firewall

DNS Record Management (PowerShell)

```
# Forward record
Add-DnsServerResourceRecordA -Name "vcenter" -ZoneName "lab.local" -IPv4Address
"192.168.1.69"

# Reverse record
Add-DnsServerResourceRecordPtr -Name "69" -ZoneName "1.168.192.in-addr.arpa" -
PtrDomainName "vcenter.lab.local"

# Verify
nslookup vcenter.lab.local
nslookup 192.168.1.69

# List all records
Get-DnsServerResourceRecord -ZoneName "lab.local"
```

Windows Certificate Utility

```
certutil -dump cert.crt
certutil -verify cert.crt
certutil -hashfile file.zip SHA256
```

---

27. Diagnostic Python Scripts

Run from Windows workstation. Requires `pip install paramiko`.

Scenario	Script
Overall health dashboard	`python quick_status.py`
NSX recovering after boot	`python nsx_monitor.py`
NSX VIP + node connectivity	`python nsx_check.py`
NSX performance + services	`python nsx_diag.py`
SDDC Manager vs NSX sync	`python sddc_nsx_status.py`
Diagnose remediate failure	`python check_remediate_error.py`
Clear stale DB locks + fix NSX status	`python clear_locks.py`
Fix stuck IN_PROGRESS tasks	`python fix_stuck_tasks.py`
Full cascade fix (all-in-one)	`python full_remediate_fix.py`
Check disconnected accounts	`python check_disconnected.py`
Update NSX password	`python nsx_cred_update.py`
NSX CPU overloaded	`python nsx_slim.py`
Restore NSX services	`python nsx_restart_all.py`
System clean after fix?	`python final_check.py`

> *All scripts are in `C:\VCF-Depot\diagnostic-scripts\`. Requires `pip install paramiko`. See `Diagnostic-Scripts-Cheatsheet.md` for full reference.*

---

28. GRUB Recovery — SDDC Manager

When all accounts are locked (root, vcf, admin):

1. Reboot VM in VMware Workstation
2. Press `e` at GRUB menu
3. Find the `linux` line, append: `init=/bin/bash`
4. Press `Ctrl+X` to boot
5. At bash prompt:

```
mount -o remount,rw /
faillock --user root --reset
faillock --user vcf --reset
reboot -f
```

> ***Note:*** *`systemctl` commands will not work in `init=/bin/bash` mode (systemd not running). Just reset faillock and reboot.*

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